

The Taylor tree commutes with parameter averaging

Claude, with Daniel Murfet

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Abstract

Astra #10 rank 1. An abstract lemma with no analytic content: if for a.e. parameter w $|Z(w) - \sum_{\alpha \in P} N^{-\alpha} (A_{\alpha}(w) \log N + B_{\alpha}(w))| \leq KR$ with a.e.-bounded, a.e.-strongly-measurable coefficients and a nonnegative integrable weight ϱ , then $|\int Z\varrho - \sum_{\alpha \in P} N^{-\alpha} ((\int A_{\alpha}\varrho) \log N + \int B_{\alpha}\varrho)| \leq KR \int \varrho$ (`integral_finite_expansion`); no separate integrability of Z is required. Specialised to a measurable family of analytic amplitudes with a common envelope, the averaged chart integral has the SAME finite Taylor tree with averaged canonical coefficients and the remainder constant multiplied by the mass of the weight (`twoD_taylor_tree_integrated`), and for a probability measure the expectation $E[Z_N]$ has the Taylor tree with expected coefficients (`twoD_taylor_tree_expectation`). Measurability in the parameter is a hypothesis here. Zero sorry/axiom.

1 The things formalised

```
theorem integral_finite_expansion {Ω : Type*} [MeasurableSpace Ω] ( : Measure Ω)
  (P : Finset )
  (Z : Ω → ) (A B : → Ω → ) ( : Ω → ) (h0 : w , 0 w) (h : Integrable )
  (hZm : AESTronglyMeasurable Z ) (hAm : P, AESTronglyMeasurable (A ) )
  (hBm : P, AESTronglyMeasurable (B ) ) (KA KB : → )
  (hA : P, w , |A w| KA ) (hB : P, w , |B w| KB )
  (K R N : )
  (hrem : w , |Z w - P, N ^ (-) * (A w * Real.log N + B w)| K * R) :
  |( w, Z w * w )
    - P, N ^ (-) * (( w, A w * w ) * Real.log N + w, B w * w )|
    K * R * w, w

theorem twoD_taylor_tree_integrated {Ω : Type*} [MeasurableSpace Ω] ( : Measure Ω)
  ... (c : Ω → × → → ) (common envelope hypotheses for every w) ( ) (h0) (h)
  (TN : ) (hN : 1 N)
  (hZm : AESTronglyMeasurable (fun w => twoDamp b N h h k k (anaAmp (c w) b)) )
  (hAm : polesBelowGen ..., AESTronglyMeasurable (fun w => canonA ... ) )
  (hBm : polesBelowGen ..., AESTronglyMeasurable (fun w => canonB ... ) ) :
  |( w, twoDamp b N h h k k (anaAmp (c w) b) * w )
    - polesBelowGen h h k k p p T,
      N ^ (-) * (( w, canonA ... * w ) * Real.log N
        + w, canonB ... * w )|
    uniformTreeConst b C L p p T h h k k D
    * (N ^ (-(2 * T)) * (1 + Real.log N)) * w, w

theorem twoD_taylor_tree_expectation ... [IsProbabilityMeasure ] ... :
  |( w, twoDamp b N h h k k (anaAmp (c w) b) )
    - polesBelowGen ..., N ^ (-) * (( w, canonA ... ) * Real.log N
      + w, canonB ... )|
```

```
uniformTreeConst b C L p p T h h k k D
* (N ^ (-(2 * T))) * (1 + Real.log N))
```

2 Mathematics

Write $E(w) = Z(w) - \sum_{\alpha} N^{-\alpha} (A_{\alpha}(w) \log N + B_{\alpha}(w))$. The a.e. bounds make E_{ϱ} , $A_{\alpha}\varrho$, $B_{\alpha}\varrho$ integrable (bounded measurable times integrable), hence Z_{ϱ} too, and linearity of the integral gives $\int Z_{\varrho} = \int E_{\varrho} + \sum_{\alpha} N^{-\alpha} (\log N \int A_{\alpha}\varrho + \int B_{\alpha}\varrho)$; finally $|\int E_{\varrho}| \leq \int |E|_{\varrho} \leq KR \int \varrho$. The specialisation feeds the uniform Taylor tree of unit 135 (remainder bound for every w) and the envelope bounds of unit 134 on the canonical coefficients (a.e. bounds, here everywhere).

3 Paper-facing meaning

Astra #10's paper-facing message: the Taylor tree is uniform on bounded analytic families (unit 135) and commutes with finite-mass averaging, so the per-stratum integral $\int Z_w(N) \rho_I(w) dw$ over the noncritical coordinates, and the expectation over a random amplitude $\eta e^{\beta s \xi_n}$ with a common envelope, have the same finite expansion with averaged/expected canonical coefficients. What is NOT yet covered: measurability of the chart integral and of the canonical coefficients in the parameter (Astra #10 ranks 2–3, via partial sums and the series formulas), and anything about $E[\log Z]$ or the fluctuation function, which need lower-tail control beyond this deterministic bound.

4 Lean notes

- `Integrable.bdd_mul` (`hg : Integrable g`) (`hf : AESTronglyMeasurable f`) (`x`, `||f x|| ≤ c`) is the a.e.-bounded-times-integrable lemma; it lets one avoid any integrability hypothesis on Z itself.
- `Finset.aestronglyMeasurable_sum` produces the Pi-sum `i s, f i`; there is no primed pointwise variant on this pin, so transfer with `.congr` and `Finset.sum_apply`.
- `norm_integral_le_of_norm_le` (`hg : Integrable g`) (`x`, `||f x|| ≤ g x`) gives $|\int f| \leq \int g$ after `← Real.norm_eq_abs`.
- For a probability measure `simp` proves `_, (1 : ℝ) = 1`.